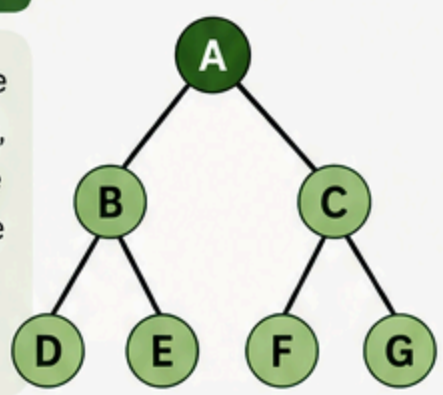
# TREES
## IN COMPUTER SYSTEMS

Trees are hierarchical data structures that organize information efficiently, enabling fast searches, insertions, and deletions.

### WHAT IS A TREE?

A tree is a non-linear structure made up of connected nodes, where there is one main node called the root and each node (except the root) has exactly one parent node.




Root
Node
Edge
Leaf

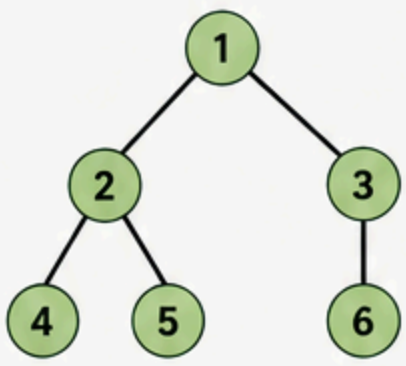
### KEY FEATURES

- Has a root.
- Each node (except the root) has exactly one parent.
- A node can have zero or more children.
- There are no cycles.

### TYPES OF TREES

#### 1. BINARY TREE

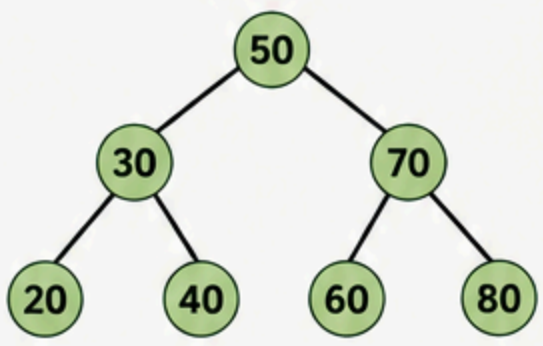
Each node has at most two children: left and right.




**Use case:** arithmetic expressions, search trees, compilers.

#### 2. BINARY SEARCH TREE (BST)

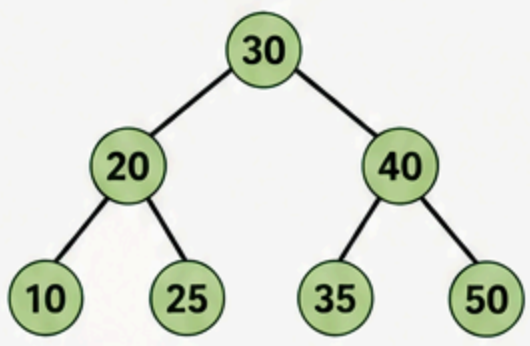
For each node, all values in the left subtree are smaller and in the right subtree are greater.




**Use case:** efficient search, sorting, databases.

#### 3. AVL TREE

A self-balanced binary search tree. The height difference between subtrees is at most 1.

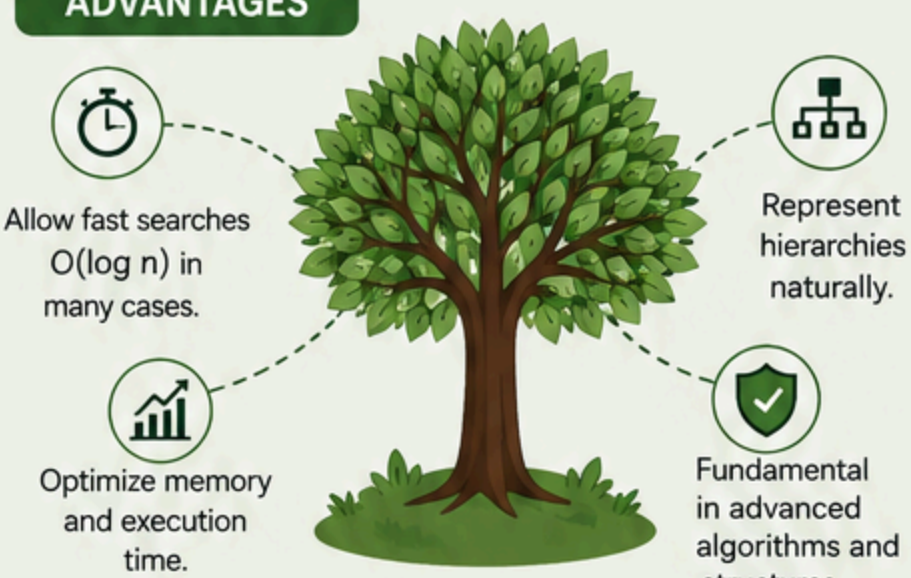



**Use case:** systems with many insertions and dynamic searches.

### APPLICATIONS IN COMPUTER SYSTEMS

- Efficient information searching.
- Indexes in databases (B-Trees, AVL).
- File systems (hierarchical directory structures).
- Compilers (syntax trees).
- Routing in networks and routing tables.
- Compression algorithms (Huffman).

### ADVANTAGES




### IN CONCLUSION

Trees are essential structures in computer science due to their efficiency in organizing and managing hierarchical data. Their use directly impacts the performance of multiple applications and systems.

**MEMBERS:** Darlin Demera
Jeffry Vera